# Capstone Project Report

(Project Term: January - May 2023)

# POULTRY FARMING IN FUTURE USING IOT

submitted by


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Project group number: KC333

Course Code: CSE445

Under the Guidance of


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**Lovely Professional University**

**Punjab, India**

**TOPIC APPROVAL PERFORMA**

School of Computer Science and Engineering (SCSE)

**Program :**P132::B.Tech. (Computer Science and Engineering)

**COURSE CODE :** CSE445 **REGULAR/BACKLOG :** Regular **GROUP NUMBER :** CSERGC0333

**Supervisor Name** : Nirmaljeet Kaur **UID :** 29395 **Designation :** Assistant Professor
**Qualification :** _____ **Research Experience :**_____

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**SPECIALIZATION AREA** : Networking and Security-I **Supervisor Signature:** _____

**PROPOSED TOPIC** : IOT Based Automated Poultry Farm for Layer Chicken

| Qualitative Assessment of Proposed Topic by PAC | | |
|---|---|---|
| **Sr.No.** | **Parameter** | **Rating (out of 10)** |
| 1 | Project Novelty: Potential of the project to create new knowledge | 5.73 |
| 2 | Project Feasibility: Project can be timely carried out in-house with low-cost and available resources in the University by the students. | 6.07 |
| 3 | Project Academic Inputs: Project topic is relevant and makes extensive use of academic inputs in UG program and serves as a culminating effort for core study area of the degree program. | 6.60 |
| 4 | Project Supervision: Project supervisor's is technically competent to guide students, resolve any issues, and impart necessary skills. | 7.60 |
| 5 | Social Applicability: Project work intends to solve a practical problem. | 6.67 |
| 6 | Future Scope: Project has potential to become basis of future research work, publication or patent. | 6.67 |

| PAC Committee Members | | |
|---|---|---|
| PAC Member (HOD/Chairperson) Name: Dr. Harwant Singh Arri | UID: 12975 | Recommended (Y/N): Yes |
| PAC Member (Allied) Name: Ravishanker | UID: 12412 | Recommended (Y/N): Yes |
| PAC Member 3 Name: Dr. Manjit Kaur | UID: 12438 | Recommended (Y/N): Yes |

**Final Topic Approved by PAC:** **IOT Based Automated Poultry Farm for Layer Chicken**

**Overall Remarks:** Approved

**PAC CHAIRPERSON Name:** 13897::Dr. Deepak Prashar **Approval Date:** 15 Mar 2023

# Students Statement

This is to signify that; we are the authors of the report. Nothing has been taken in the report directly from a different source. All information gathered from other sources has been properly referenced. We declare that we shall take full responsibility for any part of the report that are identified to have been copied. We completed the project under the guidance of Nirmaljeet Kaur, assistant professor, Lovely Professional University, Punjab, India.

Project Group Number: **KC333**


**Sadman Kamor**

Reg: **11917494**

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**Santo Saha**

Reg:**11900127**

**Snigdho Roy Bondhon**

Reg: **11919146**


Punjab, India

Date: 03-05-2023

# CERTIFICATE

This is to certify that, to the best of my knowledge and belief, the written statement made by this group of students is true. This Capstone Project was finished by them with my supervision. The current work is the outcome of their initial research, work, and study. The whole work has never been submitted for evaluation for any other degree at any university. The capstone project is acceptable for submission and fulfils some of the requirements for the award of a B. Tech degree in Computer Science and Engineering from Lovely Professional University, Punjab, India.

_____

**Nirmaljeet Kaur**

Assistant Professor,

School of Computer Science and Engineering,

Lovely Professional University

Punjab, India

Date: 03-05-2023

# ACKNOWLEDGEMENT

Any research report has to have acknowledgements because they give people a chance to express their gratitude to those who supported the project. We would want to take this opportunity to express our sincere gratitude to all those who has helped us throughout the way of completing the research in this report.

First of all, we are grateful to Nirmaljeet Kaur, our supervisor, for her essential guidance, support, and motivation during the course of the project. This research might not become achievable without her unflinching assistance and helpful suggestions.

We also would like to thank the poultry farm owners who agreed to take part in our study. Without people who volunteered to participate, this research Work would not have been completed. We value their input and show respect for all of their time and work.

We also like to express appreciation to the Lovely Professional University employees who gave us access to the tools and parameters necessary to successfully carry out this study. Their assistance and involvement were vital for this study's success.

Furthermore, we would like to thank our friends and family for their support during this research journey. We are grateful of their love and support, and their inspiration and unbreakable support have worked as an endless source of motivation.

Last but not least, we are thankful for all of our classmates for their contributions, who have given us critical feedback and suggestions on our project. We acknowledge their help and know that their opinions and recommendations were vital in developing our research.

We would like to conclude by saying thanks to everybody who has helped with our research work. We value everything you have done for us and are forever grateful for your support, encouragement, and contribution.

# ABSTRACT

The poultry industry plays a major role in the global food supply chain. It is providing a sustainable source of protein. The Food and Agriculture Organization of the United Nations (FAO) estimates that in 2020, chicken meat production reached a record high of approximately 102 million metric tons. It is approximately over 36% of global meat production. Most of the poultry farms in India are physically operational. As a result, traditional poultry farming methods face many challenges. The main challenges are labour intensive processes, environmental impact, disease management, etc. To solve these issues, automated poultry farms offer a viable option that optimizes chicken production operations. In this century, Internet of Things (IoT) technology is a very efficient way to solve the limitations. In order to increase production, efficiency, and sustainability in chicken farming, this research study examines IoT applications. In addition to discussing earlier studies and research on IoT-based poultry farming systems. This paper includes their advantages and disadvantages. The current state of IoT in poultry farming is described in the paper. The study's design and implementation of an IoT-based chicken farming system. It consists of sensors, wireless communication networks, and cloud-based data management systems, which are also covered in the report.



**Keyword:** IOT, Sensor, Arduino, Cloud

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# Introduction

The poultry farming sector makes a considerable economic contribution to many nations. For the purposes of producing eggs, meat, and feathers. With new technologies and agricultural methods emerging to increase productivity, animal care, and financial success, there have been considerable developments in poultry farming. This essay will examine numerous facets of chicken farming, including its history, diverse farming systems, production methods, difficulties, and opportunities for the future. Cloud-based data management systems are also covered in the report.

Layer chickens are known for their egg-laying capabilities. It can produce up to 300 eggs per year on average, which is equivalent to almost one egg per day. However, ensuring healthy chicken growth requires regular monitoring of various factors. The main and critical ones are temperature, air quality, humidity, ventilation, and lighting.

The automation of chicken monitoring processes can significantly contribute to the growth of the poultry trade. The management of temperature, air quality, ventilation, and other factors can be labour-intensive, and the proposed project aims to automate these processes using the Internet of Things (IoT). The system will monitor and control everything to ensure healthy chicken growth and quality egg production. The main focus of the project is the expansion of layer chickens. Overall, this project seeks to leverage technology to optimize poultry farming processes and improve the productivity and efficiency of layer chicken production. while also ensuring the health and quality of poultry products through automated monitoring and management of critical factors.

# Different Farming System

In chicken farming, there are several agricultural systems employed. Every system has its own pros and cons. Free range, cage, and barn systems are the most popular types. Free-range System: In a free-range system, the birds are given unrestricted access to food, water, and shelter when they are outdoors. This method takes a lot of space. Given that it gives birds more freedom to fly around and engage in natural activities, this method is frequently regarded as being more moral. But it also has some disadvantages. It has more susceptibility to disease and predators. It has higher production costs and increased exposure to both. Cage System: In this system, birds are kept in cages with little room for movement, frequently in large-scale operations. This system has drawn criticism because of purported problems with animal welfare. On the bright side, this method is well known for its great production and low labour costs. Barn System: Keeping birds indoors in sizable barns with access to food, water, and environmental controls is known as the "barn system." This system provides a balance between output, animal welfare, and cost effectiveness. But this system needs a substantial investment in technology and infrastructure.

# Challenges In Poultry Farming

Many difficulties face the poultry industry, including disease outbreaks, issues with food safety, and environmental effects.
Disease Outbreaks: Disease outbreaks can have a huge negative impact on the economy and public health if they affect poultry production. Many nations have implemented trade restrictions and conducted large flocking's of chickens in response to avian influenza, Newcastle disease, and other outbreaks of poultry diseases.

Food Safety Concerns: Worries about Food Safety Products made from poultry are frequently linked to foodborne illnesses, including salmonella and cyanobacteria. Antimicrobial resistance has been connected to procedures used in poultry husbandry, such as the use of antibiotics and other veterinary medications.

# Related Works

In this research paper, we observe and manage the condition of Atmosphere of a poultry farm. We are using a WI-FI module to connect the environment with the cloud database. This IOT makes a farmer smart and replaces all his hard work, which is sensible for a poultry. A system that improves the health of the poultry and increases the production of a farm. It is making a hardware system and using web technology to monitor the farm. It can check humidity, temperature, carbon dioxide, sulfur dioxide, carbon monoxide, air quality, and light sensing, which means light sensitivity. The main motto of this project is to produce low-cost hardware and a website to increase the productivity of the farm. It will reduce the cost of managing farms, decrease the efficiency of manpower, and the main part is reducing the time. The basic result is the increased production of eggs and a healthy chicken without any external hormones or medicines. The main function of this research paper is to improve the administration and monitoring of the poultry farm. [2] With the help of WI-FI module technology, the system is used by the user to improve a healthy feed for poultry, and the atmospheric conditions are controlled in the poultry farm. Even if the poultry farm owner is not on the farm, he/she can operate and control the system from any location in the world with the help of the internet. With the help of this continuing monitoring system, environmental factors like humidity, temperature, and air quality are needed for a healthy environment in a poultry farm. The main goal of this paper is to automate the control of a poultry farm through the use of WI-FI module sensors and many IOT sensors to improve the health and increase the productivity of eggs. [3] The primary goal of the system is to create an automated, environment-controlled poultry management system. Not only does the system keep an eye on the poultry coop, but it also appropriately controls these parameters. The system's strategy is to control exploitation remotely using portable handheld devices. Temperature and humidity are a crucial factor for chicken health. For taking accurate measurements of the temperature, we are using temperature sensors. We are also inculcating wireless sensor networks and general packet radio service to control and supervise the environmental factors like temperature, ammonia gas, humidity inside the farm. Food and water level are being controlled by an ultrasonic sensor. This automated system takes out a huge workload from a farmer and it narrows down the human error level to almost nothing. [4] All the components of this system are being automatically controlled by wireless sensors and wireless networks. A farmer can control and monitor all the information and components without taking a single step with his mobile devices. All the data will keep updating every minute. If the farmer wants to see the real time data, it will also be there so that farmer can supervise and regulate manually any time if necessary. [5] Poultry is one of the biggest contributors for the nutrition of a nation. It faces the

challenge of excess heat or excess cold during summer and winter. During hot summer days and cold winter nights, the health of the chickens and food quality are really in danger [6]. This study mainly focuses on cold and hot weather monitoring. The temperature is monitored and controlled by wireless sensors. By using this system, the health of the chicken and the egg off the chicken comes out in a good quality, which is great for human health [7]. The wireless system helps in achieving the production goal of the farm. The progress of the wireless sensors is focused on the study of this paper [8]. The small temperature sensor used by us has low power consumption and good sensitivity. So, we get good data with less amount of error. This is the main goal of this research.

# Present System

Almost all of the household farms in the world use the traditional technique to raise chickens for meat and egg. Everything like temperature, humidity, air quality, food, waste is being taken care of by the owner or a worker manually. It is a hard task for a person to monitor all this accurately. But it is very important to monitor all this timely and accurately. Entering the farm many times a day can also led to hygiene problems. At night if the person does not monitor the temperature, it can lead to serious health problems for the chicken. It might even cause death. With this project we are able to monitor everything with the help of sensors. The sensor will monitor the farm and in need like high or low temperature it will automatically start the cooler fan or heater. In case of bad air quality, it will turn on the exhaust fan for ventilation. After hitting the preferred temperature and air quality everything will stop automatically as well. As there is no effective system to clear the ammonia gas from the air, a person has to do it manually by turning on and off the fan [9]. The project also has lux sensors to sense day and night to turn on and off the room lights. In short, the whole farm can be managed from a phone or laptop anywhere any time with the help of IOT technology, as all the sensors are connected with IOT devices.

# Suggestive System

By monitoring, controlling, and stabilizing any kind of change in environmental factors that can be harmful to chickens, such as air quality, humidity, temperature, ventilation, automatically controlled lighting and growing poultry feed through the Internet of Things (IoT), this study aims to provide healthy chickens that produce high-quality eggs and to increase egg production in the world.

**Equipment Prerequisites**

- Arduino mega
- Power supply
- Temperature sensor
- Gas sensor
- Relay board
- LCD display
- WIFI
- LED
- Weight sensors

**Application Prerequisites**

- Arduino IDE
- Web Technologies
- Databases
- Google Firebase

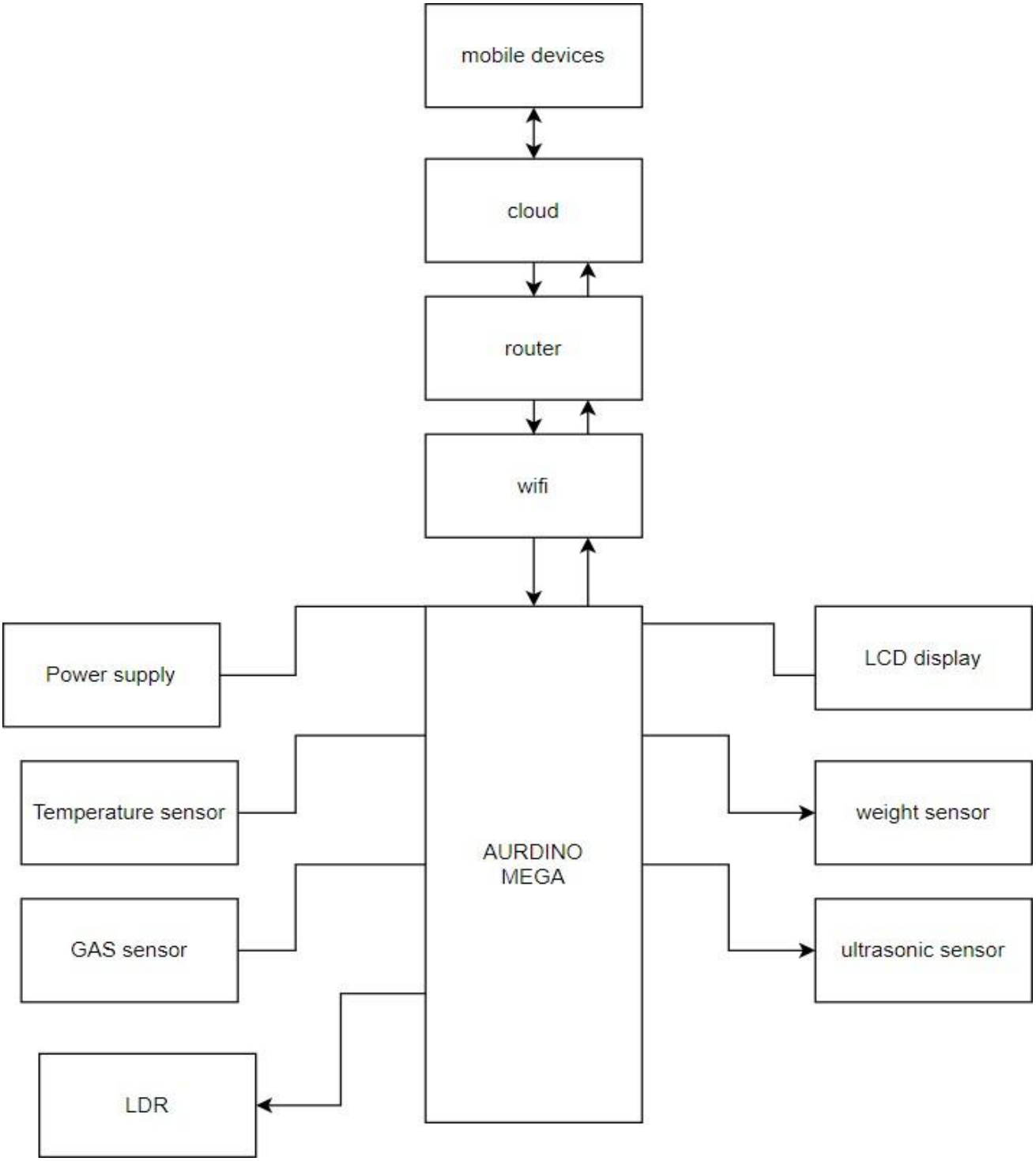



Fig.1: basic diagram of connection between the component

# The Login page code

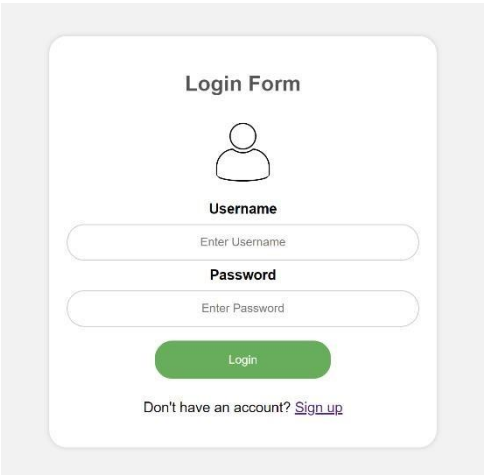
This is the login page code written in html and css and javascript.

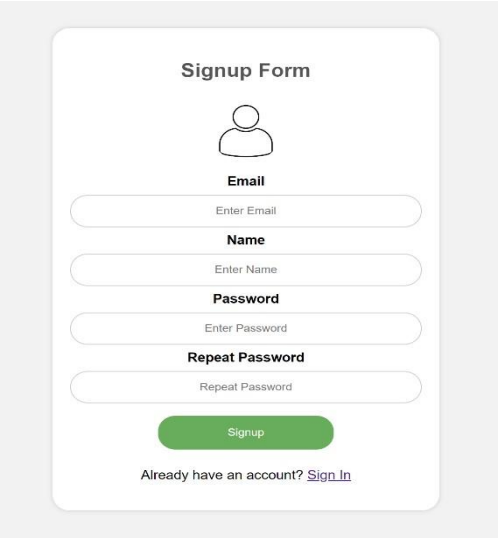
```html
<div class="container">
    <h2>Login Form</h2>
    <form action="/login" method="post" onsubmit="return validateForm()">
        <div class="imgcontainer">
            <img src="images\user-6318011_1280.png" alt="Avatar" class="avatar">
        </div>

        <label for="email"><b>Username</b></label>
        <input type="text" placeholder="Enter Username" name="email" required id="email-input">

        <label for="psw"><b>Password</b></label>
        <input type="password" placeholder="Enter Password" name="psw" required id="password-input">

        <button type="submit" id="sign-in-form">Login</button>

         <p>Don't have an account? <a href="signup.html">Sign up</a></p>

        <div class="error" id="error-message"></div>
    </form>
</div>

<script>
    function validateForm() {
  var email = document.forms[0]["email"].value;
  var password = document.forms[0]["psw"].value;

  // Check if email is valid
  var emailRegex = /^\S+@\S+\.\S+$/;
  if (!emailRegex.test(email)) {
    document.getElementById("error-message").innerHTML = "Invalid email address";
    return false;
  }

  // Check if password is at least 8 characters long
  if (password.length < 8) {
    document.getElementById("error-message").innerHTML = "Password must be at least 8 characters long";
    return false;
```

This is the authentication in google firebase which authenticate using email and the data is stored in google. This email is verify using login and if the data is not matched then is not logged.

The email also verified that means weather the email is valid or not and the password length which is greater than eight digits using java scripts valid form function.

```
<script type="module">
  // Import the functions you need from the SDKs you need
  import { initializeApp } from "https://www.gstatic.com/firebasejs/9.17.1/firebase-app.js";
  import { getAuth, signInWithEmailAndPassword } from "https://www.gstatic.com/firebasejs/9.17.1/firebase-auth.js";

  const firebaseConfig = {
    apiKey: "AIzaSyDtXPYhAXRG96ttRX-o-LG6ONY5nWooN8M",
    authDomain: "poultry-farm-0007.firebaseapp.com",
    databaseURL: "https://poultry-farm-0007-default-rtdb.firebaseio.com",
    projectId: "poultry-farm-0007",
    storageBucket: "poultry-farm-0007.appspot.com",
    messagingSenderId: "173962908871",
    appId: "1:173962908871:web:ae2c9d0f0e4d3a3ae31ae2"
  };

  // Initialize Firebase
 const app = initializeApp(firebaseConfig);


const auth = getAuth(app);

const emailInput = document.getElementById('email-input');
const passwordInput = document.getElementById('password-input');
const signIpForm = document.getElementById('sign-in-form');

signIpForm.addEventListener('click', (event) => {
  event.preventDefault();
  const email = emailInput.value;
  const password = passwordInput.value;

 signInWithEmailAndPassword(auth, email, password)
  .then((userCredential) => {
    // Signed in
    const user = userCredential.user;
    window.location.href = "main menu.html";

```



# The Signup page code

This is the code for signup form using html, css and javascript.

```
signup.html > {} "signup.html" > html > body > div.container > form#signup-form > p > a
<div class="container">
    <h2>Signup Form</h2>
    <form id="signup-form" action="/signup" method="post" onsubmit="return validateForm()">
        <div class="imgcontainer">
            <img src="images\user-6318011_1280.png" alt="Avatar" class="avatar">
        </div>

        <label for="email"><b>Email</b></label>
        <input type="email" placeholder="Enter Email" name="email" required id="email-input">

        <label for="name"><b>Name</b></label>
        <input type="text" placeholder="Enter Name" name="name" required>

        <label for="psw"><b>Password</b></label>
        <input type="password" placeholder="Enter Password" name="psw" required id="password-input">

        <label for="psw-repeat"><b>Repeat Password</b></label>
        <input type="password" placeholder="Repeat Password" name="psw-repeat" required>

        <button type="submit" id="sign-up-form">Signup</button>
        <p>Already have an account? <a href="login.html">Sign In</a></p>

        <div class="error" id="error-message"></div>
    </form>
</div>
<script type="text/javascript">
function validateForm() {
  var email = document.forms[0]["email"].value;
  var username = document.forms[0]["name"].value;
  var password = document.forms[0]["psw"].value;
```

The signup form collects the data of email, name and password and it stores in google firebase. We use two times password for correct verification. If any password is wrong then that means the password is not matched then the password is not stored means the signup form is not done. The email is verified by google if you enter wrong email then it will not collect the data from this form.

```
signup.html
signup.html > {} "signup.html" > html > body > div.container > form#signup-form > p > a
      document.getElementById("error-message").innerHTML = "Invalid email address";
      return false;
    }



    // Check if password is valid
    if (password.length < 8) {
      document.getElementById("error-message").innerHTML = "Password must be at least 8 characters long";
      return false;
    }

    // Check if passwords match
    if (password != repeatPassword) {
      document.getElementById("error-message").innerHTML = "Passwords do not match";
      return false;
    }

    return true;
  }

  </script>

  <script type="module">
    // Import the functions you need from the SDKs you need
    import { initializeApp } from "https://www.gstatic.com/firebasejs/9.17.1/firebase-app.js";
    import { getAuth, createUserWithEmailAndPassword } from "https://www.gstatic.com/firebasejs/9.17.1/firebase-auth.js";

    const firebaseConfig = {
      apiKey: "AIzaSyDtXPYhAXRG96ttRX-o-LG6ONY5nWooN8M",
      authDomain: "poultry-farm-0007.firebaseapp.com",
```

In this code the data is send to the google firebase using authentication credentials like apikey, authentication, projectId, storageBucket, appId.

```
signup.html > {} "signup.html" > html > body > div.container > form#signup-form > p > a
  // Import the functions you need from the SDKs you need
  import { initializeApp } from "https://www.gstatic.com/firebasejs/9.17.1/firebase-app.js";
  import { getAuth, createUserWithEmailAndPassword } from "https://www.gstatic.com/firebasejs/9.17.1/firebase-auth.js";

  const firebaseConfig = {
    apiKey: "AIzaSyDtXPYhAXRG96ttRX-o-LG6ONY5nWooN8M",
    authDomain: "poultry-farm-0007.firebaseapp.com",
    databaseURL: "https://poultry-farm-0007-default-rtdb.firebaseio.com",
    projectId: "poultry-farm-0007",
    storageBucket: "poultry-farm-0007.appspot.com",
    messagingSenderId: "173962908871",
    appId: "1:173962908871:web:ae2c9d0f0e4d3a3ae31ae2"
  };

  // Initialize Firebase
 const app = initializeApp(firebaseConfig);


const auth = getAuth(app);

const emailInput = document.getElementById('email-input');
const passwordInput = document.getElementById('password-input');
const signUpForm = document.getElementById('sign-up-form');

signUpForm.addEventListener('click', (event) => {
  event.preventDefault();
  const email = emailInput.value;
  const password = passwordInput.value;

 createUserWithEmailAndPassword(auth, email, password)
  .then((userCredential) => {
```

# The Main Menu page code

The main menu page is containing all the sensor data along with about page and contact page. The logout button is also using for logging out the current user. The sensor data like temprature, humidity, CO2 concentration, CO concentration, NH3 concentration, Smoke presence or not is monitoring. The last two button is showing weather the light is on or off in the farm and the weight of the garbage is also shown from this button.

All the data is coming from the Think Speak server where the data is uploading from the hardware components connected with the Arduino mega. And the data is collecting  real-time data means whenever the physical hardware component is able to connect to the think speak server, then the data will automatically upload in this page using javascript and the think speak read link.

main menu.html

main menu.html > {} "main%20menu.html"

```html
</div>
<div class="buttons">
  <a href="#" id="button8">
    Light <span class="hover-text">This button shows the Methane level</span>
  </a>
  <a href="#" id="button8">
    Garbage <span class="hover-text" id="garbage_text">This button shows the Methane level</span>
  </a>
</div>




<footer>
  <div class="footer-content">
    <h3>POULTRY FARMING IN FUTURE USING IOT</h3>
    <p>This is a website where you can check your farm temprature,humidity like many more thing whish is essential for
    <ul class="socials">
      <li><a href="#"><i class="fa fa-facebook"></i></a></li>
      <li><a href="#"><i class="fa fa-twitter"></i></a></li>
      <li><a href="#"><i class="fa fa-instagram"></i></a></li>
    </ul>
  </div>
  <div class="footer-bottom">
    <p>All rights reserved &copy; POULTRY FARMING IN FUTURE USING IOT 2023</p>
  </div>
</footer>
```

main menu.html

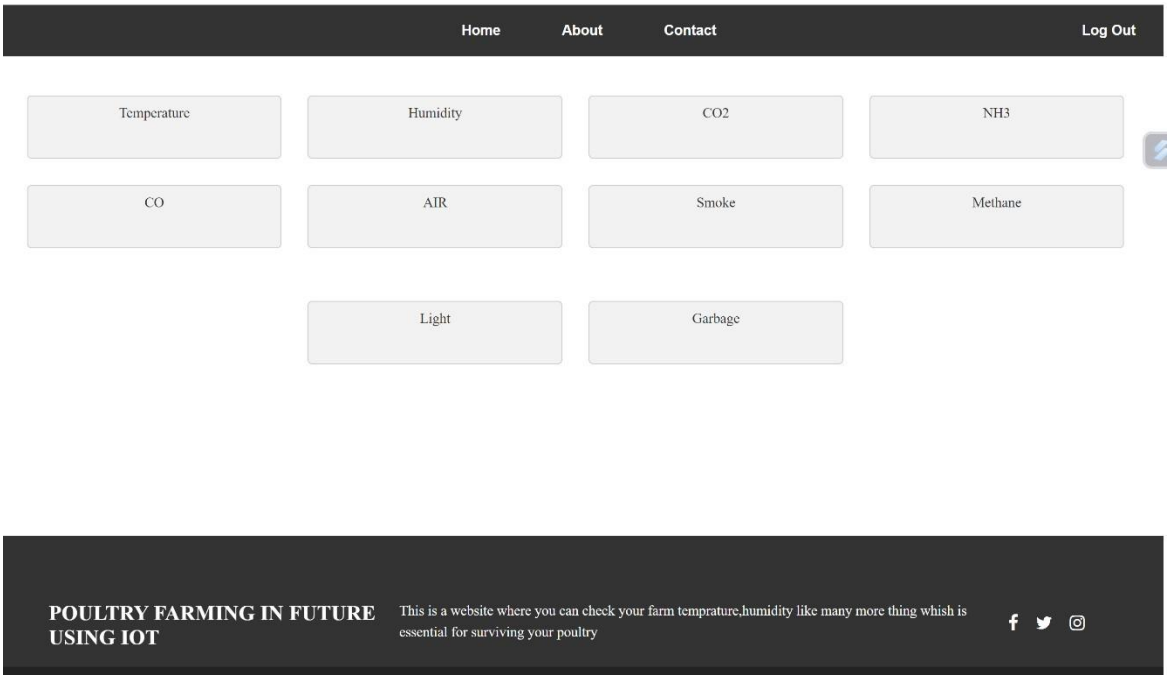
main menu.html > {} "main%20menu.html"

```javascript
<script type="text/javascript">
const button1 = document.getElementById("button1");
const button2 = document.getElementById("button2");
const button3 = document.getElementById("button3");
const button4 = document.getElementById("button4");
const button5 = document.getElementById("button5");
const button6 = document.getElementById("button6");
const button7= document.getElementById("button7");
const button8 = document.getElementById("button8");


button1.addEventListener("click", function() {
  //alert("Button 1 was clicked!");
  window.location.href = "temprature.html";
});

button2.addEventListener("click", function() {
  //alert("Button 2 was clicked!");
  window.location.href = "humidity.html";
});
button3.addEventListener("click", function() {
  //alert("Button 3 was clicked!");
  window.location.href = "CO2.html";
});
button4.addEventListener("click", function() {
  //alert("Button 4 was clicked!");
  window.location.href = "NH3.html";
});
button5.addEventListener("click", function() {
  //alert("Button 5 was clicked!");
  window.location.href = "CO.html";
```

main menu.html

main menu.html > {} "main%20menu.html"

```
import { initializeApp } from "https://www.gstatic.com/firebasejs/9.17.1/firebase-app.js";

import { getAuth, signOut } from "https://www.gstatic.com/firebasejs/9.17.1/firebase-auth.js";
const firebaseConfig = {
  apiKey: "AIzaSyDtXPYhAXRG96ttRX-o-LG6ONY5nWooN8M",
  authDomain: "poultry-farm-0007.firebaseapp.com",
  databaseURL: "https://poultry-farm-0007-default-rtdb.firebaseio.com",
  projectId: "poultry-farm-0007",
  storageBucket: "poultry-farm-0007.appspot.com",
  messagingSenderId: "173962908871",
  appId: "1:173962908871:web:ae2c9d0f0e4d3a3ae31ae2"
};

// Initialize Firebase
const app = initializeApp(firebaseConfig);


const auth = getAuth(app);

const logout = document.getElementById('logout');

logout.addEventListener('click', (event) => {


  signOut(auth).then(() => {
    window.location.href = "login.html";
  }).catch((error) => {
    // An error happened.
  });
```



# The Sensors Graph Code

The sensor graph code page is collecting each data from think speak field which is various for all the sensors. The graph had done by chart.js which is a java script file and responsible for making this chart.

The data which is collect from think speak server at first set into local storage of a webserver and then the data is collecting from the local storage, and it shown in the page and the data is auto refresh in every five seconds. For this reason, every time whenever the data is uploading in think speak server it will be auto refresh.

The time from think speak server is based on UTC and we are converting the UTC time to IST that's why the time which shown in graph is IST time.

The graph type is bar type, and we can change it in various way whatever we want. The most visible graph is bar graph that is why we use this type.

Every graph page has a print option where the user can print the data.

```
temprature.html > ...
<script>
 // Define the chart data

 var url= "https://api.thingspeak.com/channels/2039523/fields/1.json?results=20";
 var time=[];
 var tempp=[];
 fetch(url)
 .then(response => response.json())
 .then(data => {
   // Parse data and create HTML

   for (var i = 0; i < 20; i++) {

       // Extract the hour and minute values from the GMT time string
       const hour = parseInt(data.feeds[i].created_at.substring(11, 13));
       const minute = parseInt(data.feeds[i].created_at.substring(14, 16));

       // Add 5 hours and 30 minutes to the hour and minute values
       let istHour = hour + 5;
       let istMinute = minute + 30;

       // Adjust the hour and minute values if necessary
       if (istMinute >= 60) {
         istHour += 1;
         istMinute -= 60;
       }
       if (istHour >= 24) {
         istHour -= 24;
       }

       // Convert the IST hour and minute values back to strings, with leading zeros if necessary
       const istHourStr = istHour.toString().padStart(2, '0');
0
Ln 1, Col 1   Spaces: 2   UTF-8   CRLF   HTML   Go Live
```

temprature.html - capstone website - Visual Studio Code

temprature.html

```
            localStorage.setItem('temprature',tempp);
            localStorage.setItem('time',time);
        })
        .catch(error => console.error(error));

    </script>

    <script type="text/javascript">

        var timee=localStorage.getItem('time');
        var temp=localStorage.getItem('temprature');
        timee=timee.split(',');
        temp=temp.split(',');




        var data = {
            labels: timee,
            datasets: [{
                label: "Temprature",
                data: temp,
                backgroundColor: [
                        'rgba(255, 99, 132, 0.2)',
                        'rgba(54, 162, 235, 0.2)',
                        'rgba(255, 206, 86, 0.2)',
                        'rgba(75, 192, 192, 0.2)',
                        'rgba(153, 102, 255, 0.2)',
                        'rgba(255, 159, 64, 0.2)'
                    ],
                    borderColor: [
```

temprature.html - capstone website - Visual Studio Code

temprature.html

```
                fill: false
            }]
        };

        // Create a new Chart.js chart
        var ctx = document.getElementById("my-chart").getContext("2d");
        var myChart = new Chart(ctx, {
            type: "bar",
            data: data,
            options: {
                scales: {
                    yAxes: [{
                        ticks: {
                            beginAtZero: true
                        }
                    }]
                }
            }
        });

        setInterval(function() {
            location.reload();
    }, 5000);
    </script>
</body>
</html>
```

The above graph is temperature graph where it is measure in temperature (degree centigrade) in the Y axis and the time is measure in the X axis.

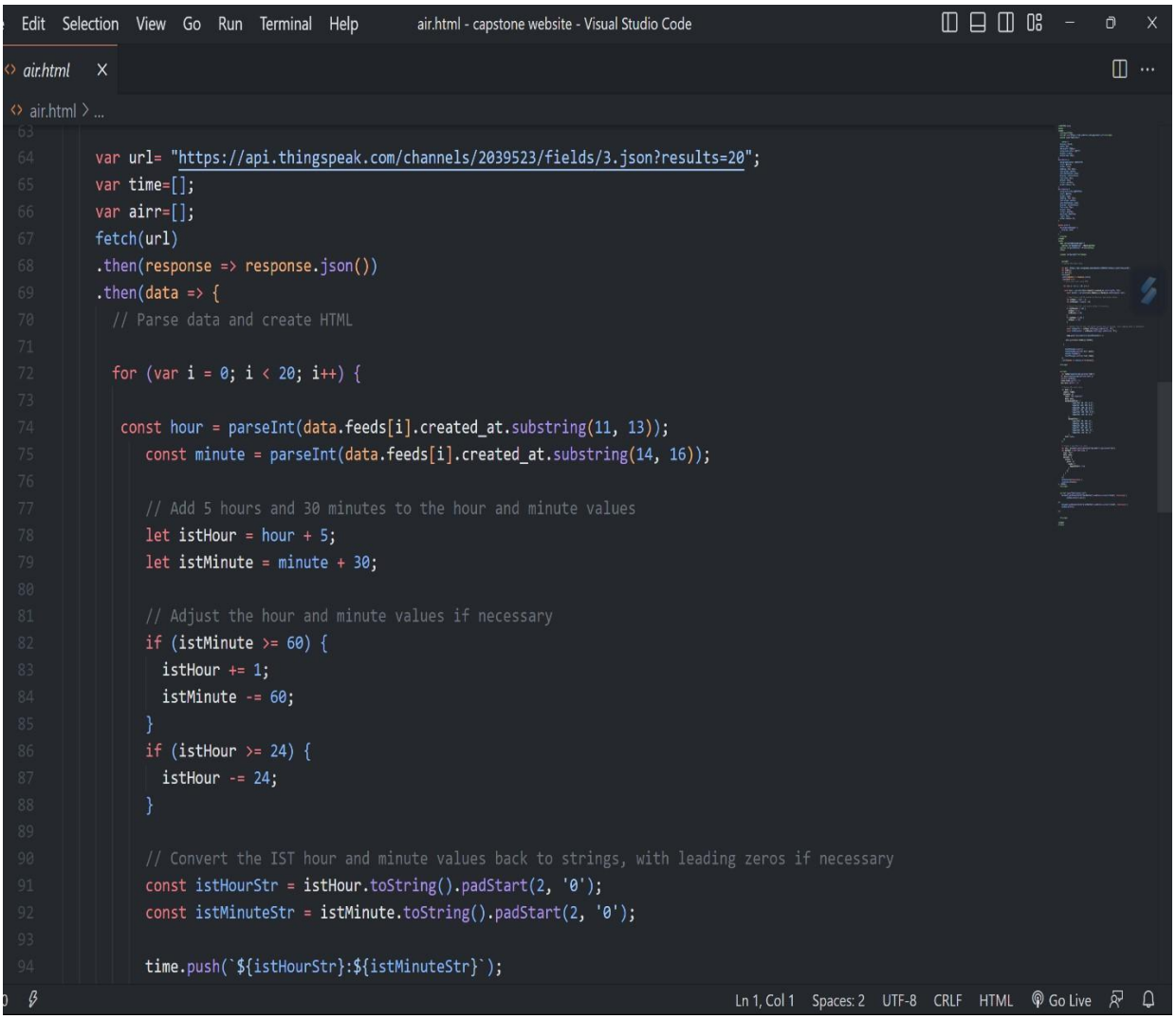
humidity.html X

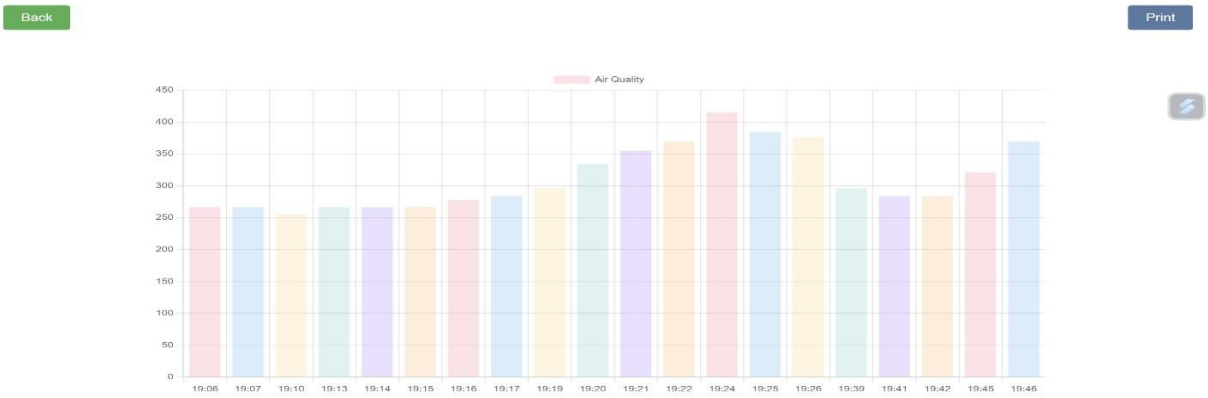
humidity.html > ...

```
var url= "https://api.thingspeak.com/channels/2039523/fields/2.json?results=20";
var time=[];
var humid=[];
fetch(url)
.then(response => response.json())
.then(data => {
  // Parse data and create HTML

  for (var i = 0; i < 20; i++) {

   const hour = parseInt(data.feeds[i].created_at.substring(11, 13));
      const minute = parseInt(data.feeds[i].created_at.substring(14, 16));

      // Add 5 hours and 30 minutes to the hour and minute values
      let istHour = hour + 5;
      let istMinute = minute + 30;

      // Adjust the hour and minute values if necessary
      if (istMinute >= 60) {
        istHour += 1;
        istMinute -= 60;
      }
      if (istHour >= 24) {
        istHour -= 24;
      }

      // Convert the IST hour and minute values back to strings, with leading zeros if necessary
      const istHourStr = istHour.toString().padStart(2, '0');
      const istMinuteStr = istMinute.toString().padStart(2, '0');

      time.push(`${istHourStr}:${istMinuteStr}`);
```

Ln 1, Col 1 Spaces: 2 UTF-8 CRLF HTML Go Live

The above graph is humidity graph where it is measure in humidity in the Y axis and the time is measure in the X axis.






The above graph is air quality graph where it is measure in air quality in the Y axis and the time is measure in the X axis.

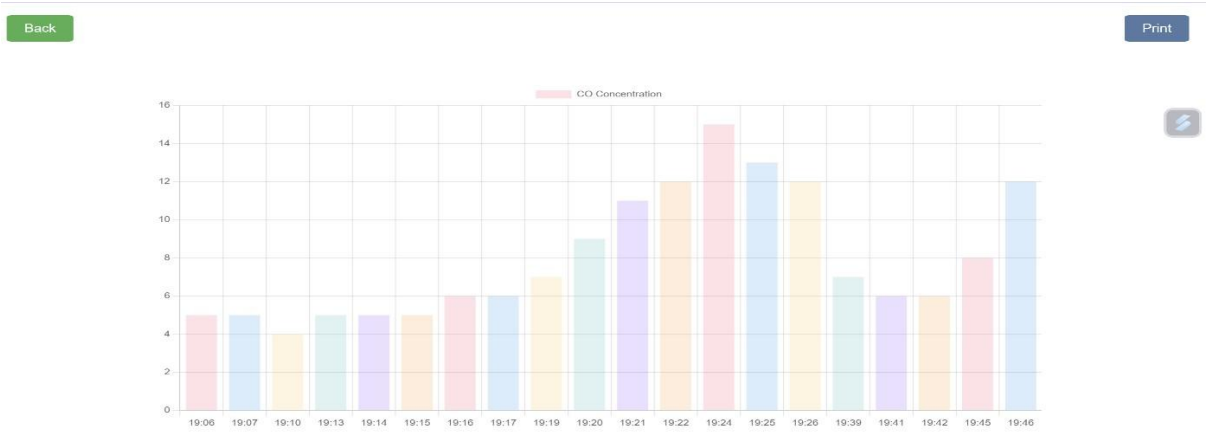
Terminal Help CO.html - capstone website - Visual Studio Code

CO.html

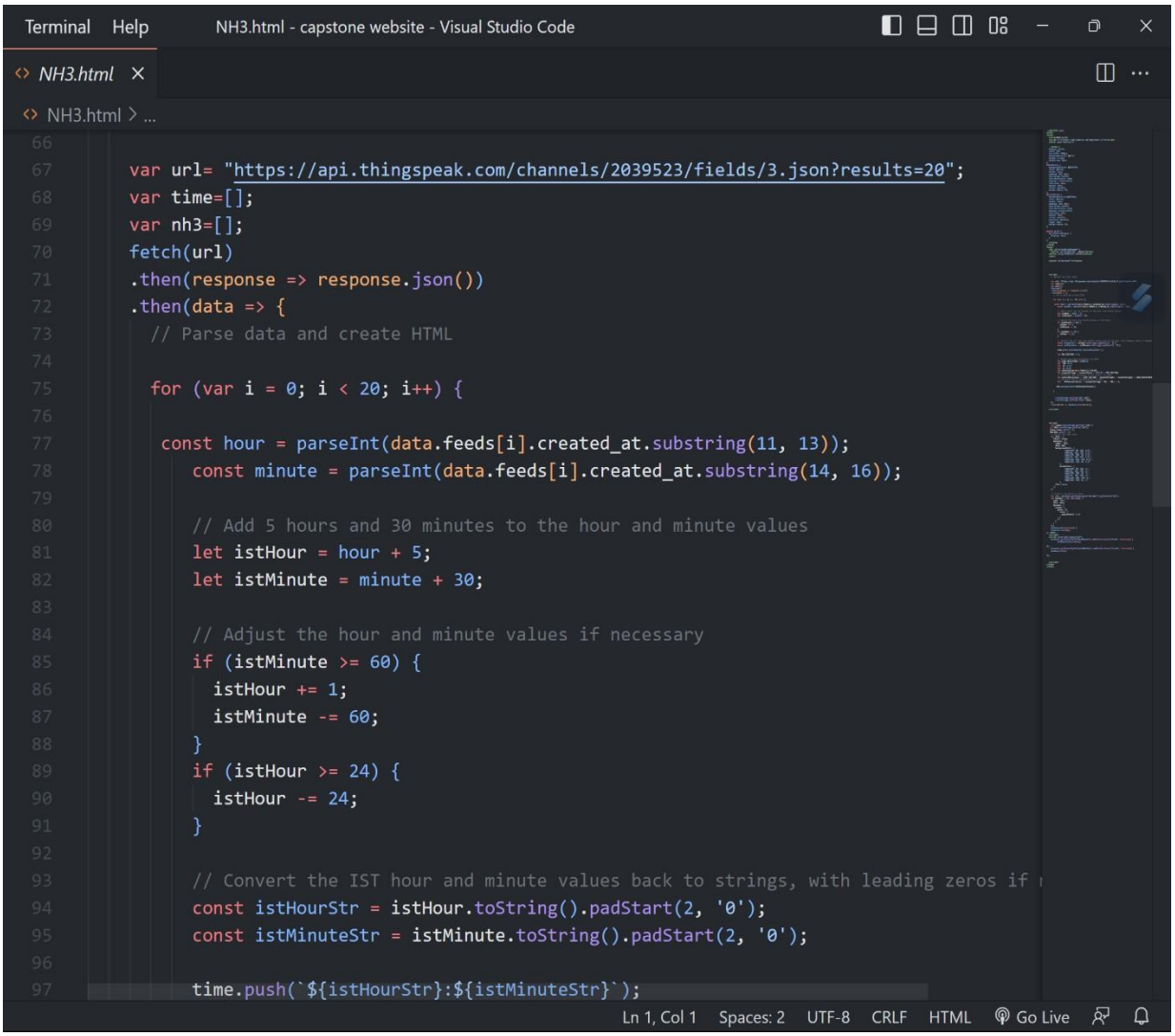
CO.html > ...

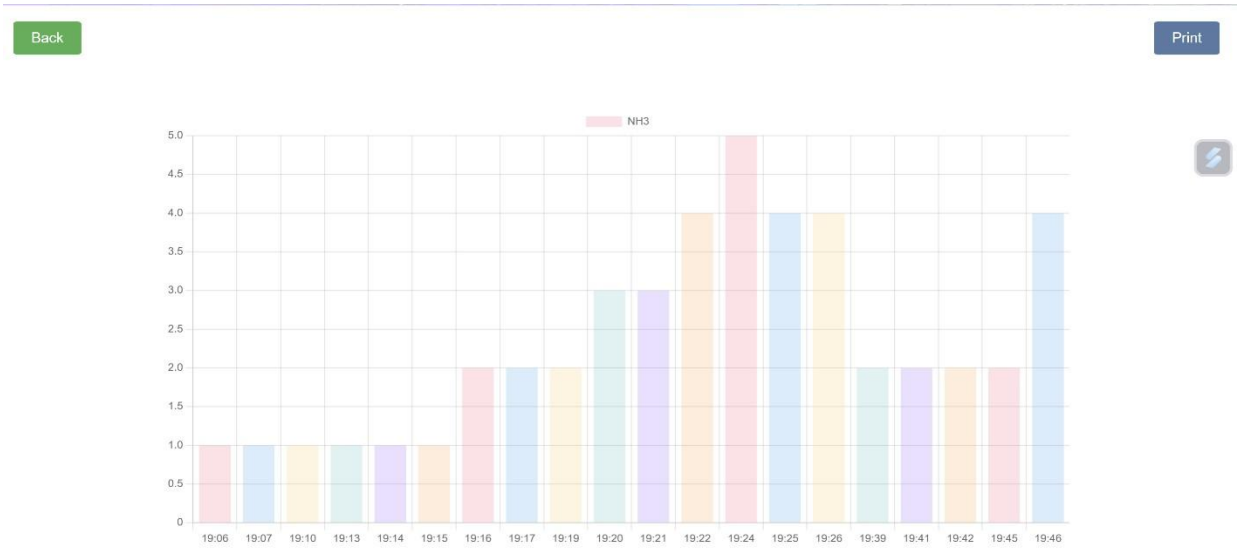
```
var url= "https://api.thingspeak.com/channels/2039523/fields/3.json?results=20";
var time=[];
var co_value=[];
fetch(url)
.then(response => response.json())
.then(data => {
  // Parse data and create HTML

  for (var i = 0; i < 20; i++) {

   const hour = parseInt(data.feeds[i].created_at.substring(11, 13));
      const minute = parseInt(data.feeds[i].created_at.substring(14, 16));

      // Add 5 hours and 30 minutes to the hour and minute values
      let istHour = hour + 5;
      let istMinute = minute + 30;

      // Adjust the hour and minute values if necessary
      if (istMinute >= 60) {
        istHour += 1;
        istMinute -= 60;
      }
      if (istHour >= 24) {
        istHour -= 24;
      }

      // Convert the IST hour and minute values back to strings, with leading zeros if
      const istHourStr = istHour.toString().padStart(2, '0');
      const istMinuteStr = istMinute.toString().padStart(2, '0');

      time.push(`${istHourStr}:${istMinuteStr}`);
```

Ln 1, Col 1 Spaces: 2 UTF-8 CRLF HTML Go Live

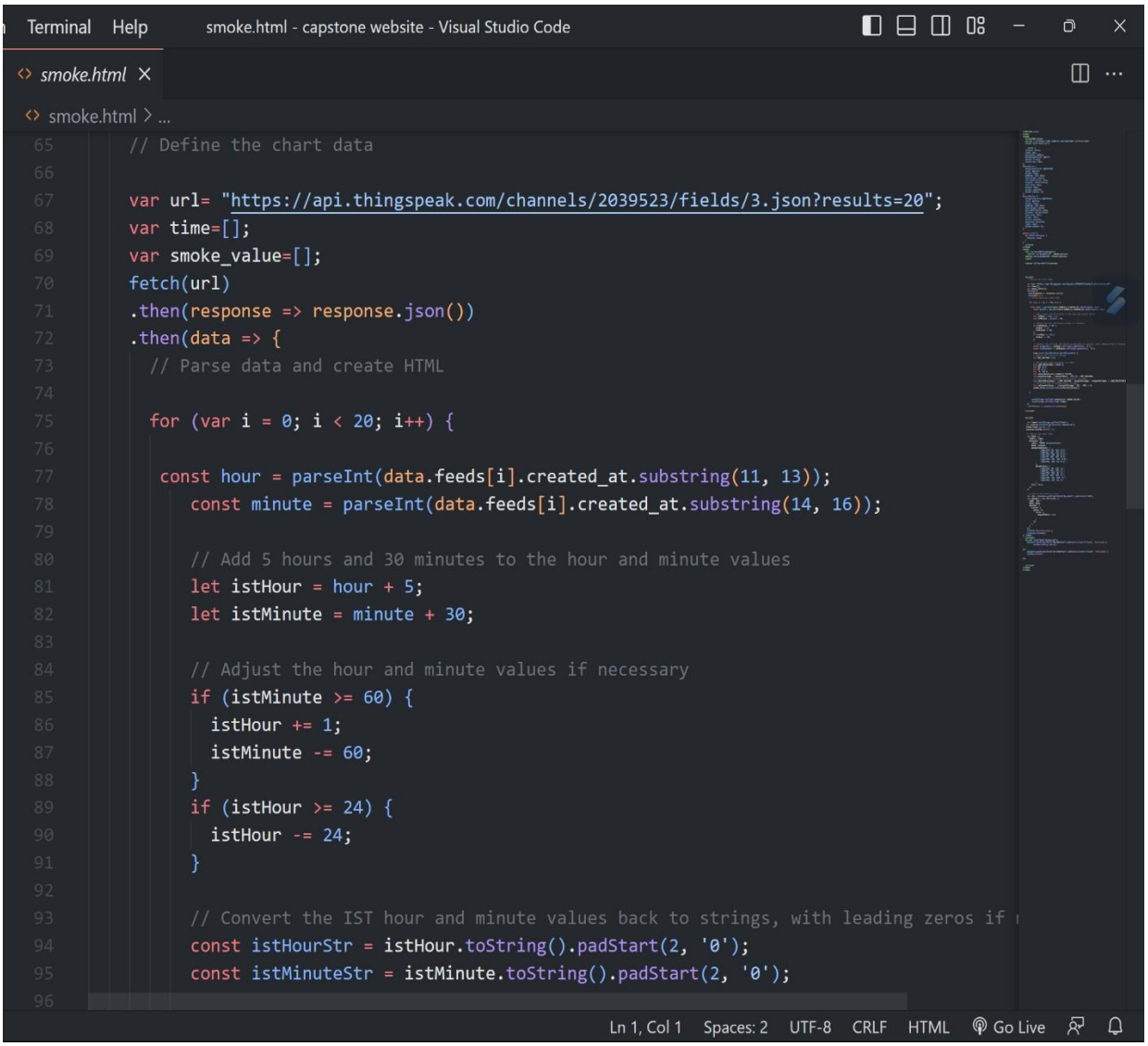



The above graph is co concentration graph where it is measure in co concentration in the Y axis and the time is measure in the X axis.

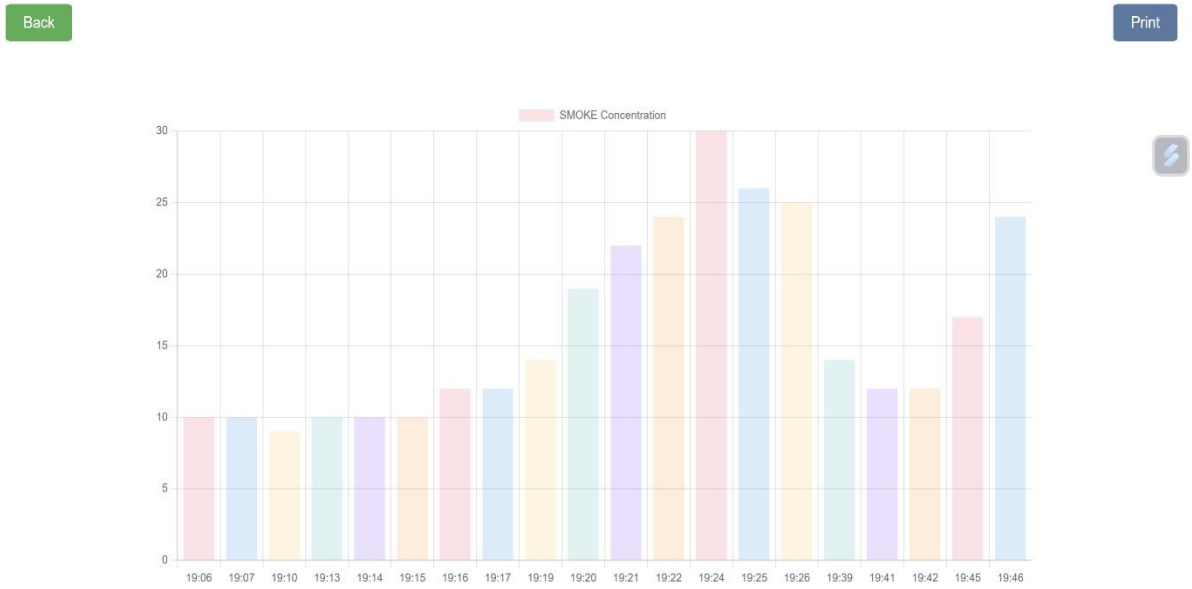




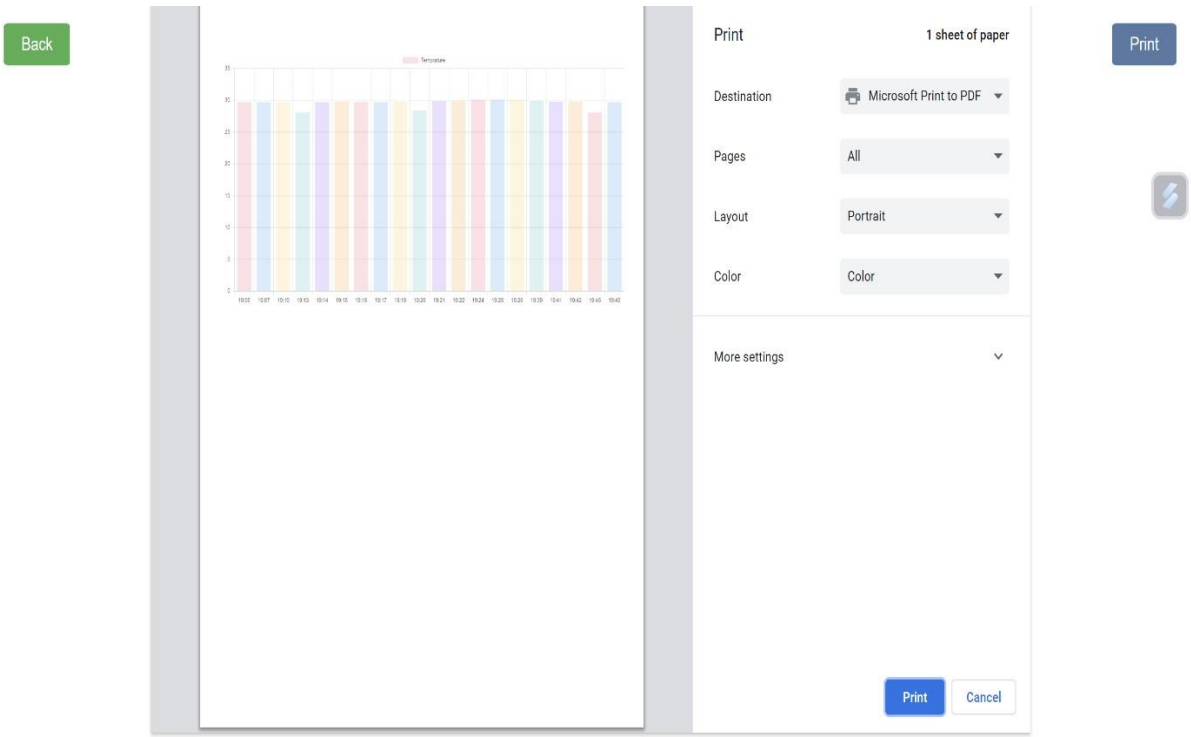

The above graph is NH3 graph where it is measure in nh3 concentration in the Y axis and the time is measure in the X axis.




```
// Define the chart data

var url= "https://api.thingspeak.com/channels/2039523/fields/3.json?results=20";
var time=[];
var smoke_value=[];
fetch(url)
.then(response => response.json())
.then(data => {
  // Parse data and create HTML

  for (var i = 0; i < 20; i++) {

   const hour = parseInt(data.feeds[i].created_at.substring(11, 13));
      const minute = parseInt(data.feeds[i].created_at.substring(14, 16));

      // Add 5 hours and 30 minutes to the hour and minute values
      let istHour = hour + 5;
      let istMinute = minute + 30;

      // Adjust the hour and minute values if necessary
      if (istMinute >= 60) {
        istHour += 1;
        istMinute -= 60;
      }
      if (istHour >= 24) {
        istHour -= 24;
      }

      // Convert the IST hour and minute values back to strings, with leading zeros if
      const istHourStr = istHour.toString().padStart(2, '0');
      const istMinuteStr = istMinute.toString().padStart(2, '0');
```




The above graph is smoke concentration graph where it is measure in smoke concentration in the Y axis and the time is measure in the X axis.




This is after the click of the print option from any other graph, and it can print in any format.

# The Arduino Code

capstone.ino

```
#include <SoftwareSerial.h>
#include <LiquidCrystal.h>
#include <MQ135.h>
#include <DHT.h>
#define DHTPIN 5      // what pin we're connected to
#define DHTTYPE DHT11    // DHT 11
DHT dht(DHTPIN, DHTTYPE);

const int rs = 8, en = 9, d4 = 10, d5 = 11, d6 = 12, d7 = 13;
LiquidCrystal lcd(rs, en, d4, d5, d6, d7);
#define RX 50
#define TX 51
int  sensorIn = A0;

int time=0;
 int distance=0;
 int trig=48;
 int echo=49;

SoftwareSerial esp8266(RX,TX);
String AP = "pompom";         // AP NAME
String PASS = "9647937614"; // AP PASSWORD 8918735480
String API = "KKOVUVBR72KUVXM0";    // Write API KEY
String HOST = "api.thingspeak.com";
String PORT = "80";
int countTimeCommand;
boolean found = false;
void setup() {
  // put your setup code here, to run once:
  lcd.begin(16, 4);
  dht.begin();
  pinMode(53,INPUT);
  pinMode(echo,INPUT);
  pinMode(trig,OUTPUT);
  pinMode(46,OUTPUT);
```

capstone.ino

```
  pinMode(46,OUTPUT);
  Serial.begin(9600);
  esp8266.begin(115200);
  sendCommand1("AT",5,"OK");
  sendCommand1("AT+RST",5,"OK");
  sendCommand1("AT+CWMODE=3",5,"OK");
  sendCommand1("AT+CWJAP=\""+ AP +"\",\""+ PASS +"\"",20,"OK");
  lcd.clear();
  lcd.setCursor(2, 0);
        lcd.print("WIFI Connect");
        lcd.setCursor(2, 2);
        lcd.print("READY TO GET");
        lcd.setCursor(6, 3);
        lcd.print("DATA");
        delay(3000);

        lcd.setCursor(0, 0);
        lcd.print("     Start     ");
        for(int i=5;i>=0;i--){
          lcd.setCursor(0, 2);
          lcd.print("                ");
          lcd.setCursor(7, 2);
          lcd.print(i);
          lcd.setCursor(0, 3);
          lcd.print("                ");
          delay(500);
        }


}

void loop() {
  // put your main code here, to run repeatedly:
  String getData1 = "GET /update?api_key="+ API +"&field1="+getTemperatureValue()+
  sendCommand("AT+CIPMUX=1",5,"OK");
```

apstone.ino

```
  sendCommand("AT+CIPSTART=0,\"TCP\",\""+ HOST +"\","+ PORT,10,"OK");
  sendCommand("AT+CIPSEND=0," +String(getData1.length()+4),10,">");
  esp8266.println(getData1);
  sendCommand("AT+CIPCLOSE=0",5,"OK");

}


String getultrasonic(){
  digitalWrite(trig,HIGH);
  delayMicroseconds(10);
  digitalWrite(trig,LOW);
  time=pulseIn(echo,HIGH);
  distance=(time*0.034)/2;
  Serial.print("Distance is : ");
  Serial.println(distance);
  lcd.setCursor(0,3);
  lcd.print("                    ");
  lcd.setCursor(0,3);

  lcd.print("Garbage is full");
  delay(5000);
  return String(distance);


}

String getldrvalue(){
  int data=digitalRead(53);
  Serial.print("light sensor value is ");
  Serial.println(data);
  lcd.setCursor(0,3);
  lcd.print("                    ");
  lcd.setCursor(1,3);

  if(data>0){
```

apstone.ino

```
  lcd.setCursor(1,3);

  if(data>0){
    digitalWrite(46,HIGH);
    lcd.print("Light is ON");
  }
  else if(data<1){
    digitalWrite(46,LOW);
    lcd.print("Light is OFF");
  }

  delay(5000);
  return String(data);
}




String getTemperatureValue(){

  float t=dht.readTemperature();
 lcd.setCursor(0,0);
 lcd.print("Temprature:");
  lcd.print(t);
  delay(50);
  return String(t);




}
String getHumidityValue(){

  float h = dht.readHumidity();
  lcd.setCursor(0,1);
```

capstone.ino

```
}
String getHumidityValue(){

  float h = dht.readHumidity();
  lcd.setCursor(0,1);
  lcd.print("Humidity: ");
  lcd.print(h);
  lcd.print("%");
  delay(50);
  return String(h);

}


String getAirValue(){
  MQ135 gasSensor = MQ135(A0);
 float air_quality = gasSensor.getPPM();
   lcd.setCursor(0,2);
   lcd.print("AirQuality:");
   lcd.print(air_quality);
   lcd.print("IQ%");
   return String(air_quality);
}

void sendCommand(String command, int maxTime, char readReplay[]) {
  Serial.println(command);
  while(countTimeCommand < (maxTime*1))
  {
    esp8266.println(command);//at+cipsend

    if(esp8266.find(readReplay))//ok
    {
      found = true;
      break;
```

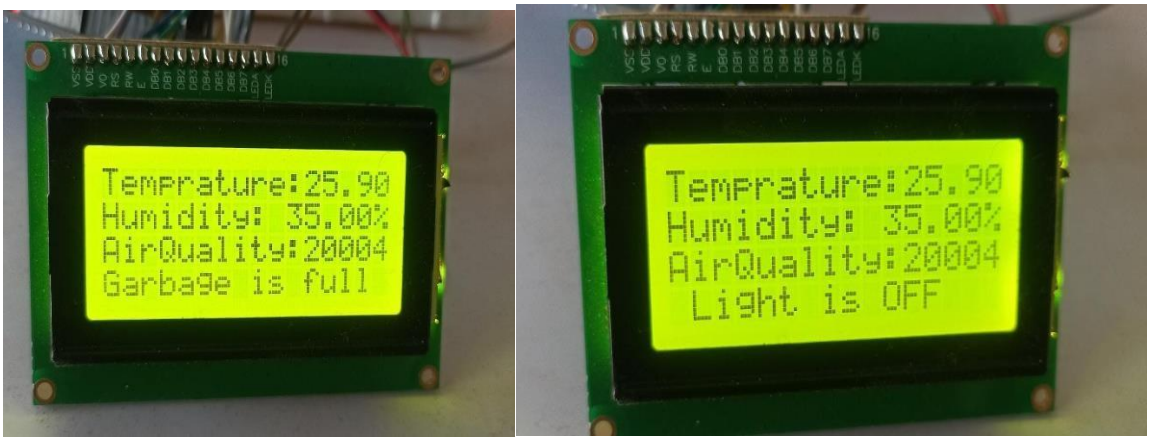
capstone.ino

```
    esp8266.println(command);//at+cipsend

    if(esp8266.find(readReplay))//ok
    {
      found = true;
      break;
    }


    countTimeCommand++;
  }

  if(found == true)
  {
    Serial.println("Connection OK");

    countTimeCommand = 0;
  }

  if(found == false)
  {
    Serial.println("Unable to connect");
    countTimeCommand = 0;

  }

  found = false;
 }

void sendCommand1(String command, int maxTime, char readReplay[]) {
  Serial.println(command);
  while(countTimeCommand < (maxTime*1))
  {
    esp8266.println(command);//at+cipsend

```

capstone.ino

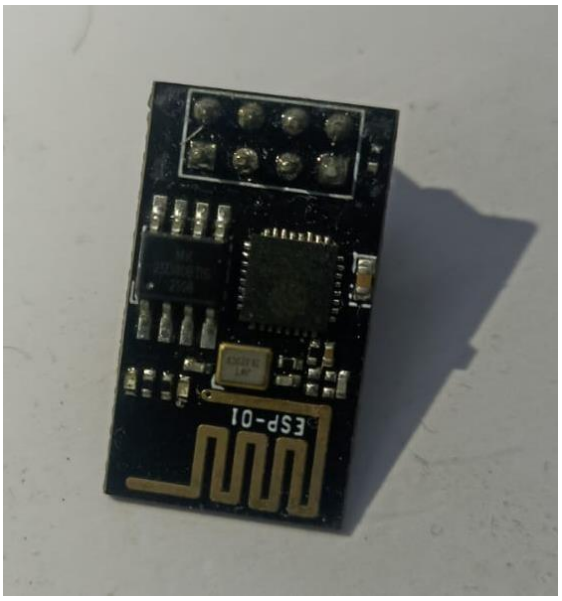
```
      if(esp8266.find(readReplay))//ok
      {
        found = true;
        break;
      }


      countTimeCommand++;
    }

    if(found == true)
    {
      Serial.println("Connection OK");
      lcd.setCursor(3,2);
      lcd.print("Connecting");
      delay(2000);
      countTimeCommand = 0;
    }

    if(found == false)
    {
      Serial.println("Unable to connect");
      countTimeCommand = 0;
      lcd.setCursor(0,1);
      lcd.print("Connection Fail");
      lcd.setCursor(1,2);
      lcd.print("Reconnecting");
      delay(3000);


    }

    found = false;
  }
```

# Modules Description



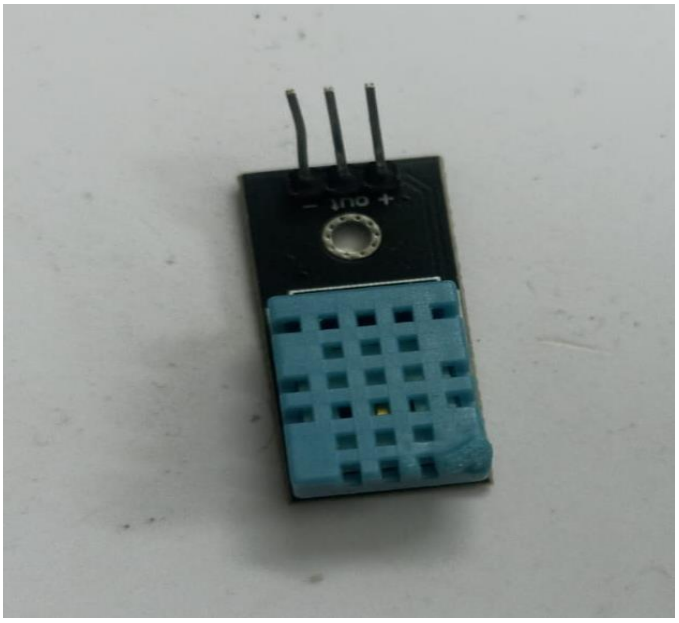

LCD 16*4

LCD 16*4 is a liquid crystal display which is used to display the sensor data. It have sixteen column and four row where you can print any word from “A-Z”, “a-z”, “0-9” in any single units. Total 64 characters is displayed in a same time. It is connected through any microcontroller like Arduino , NodeMCU, raspberrypi. It has 16 pins and the 12 pins are used to print the data and the rest four pins is not using .



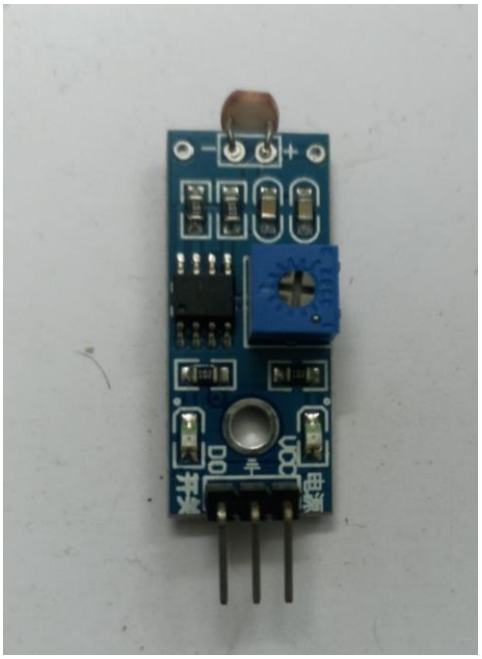

ESP8266-01 Wi-Fi module.

ESP8266-01-this is a low-cost Wi-Fi microchip.it provides Wi-Fi connectivity. I allow to connect with wireless networks with RX and Tx pin. It establishes communications with another device connected with internet is a receiver and TX is a transmit pin. This module uses AT command to configure its functions.it operates at 3.3 volts.it has 512kb or 1MB of flash memory.



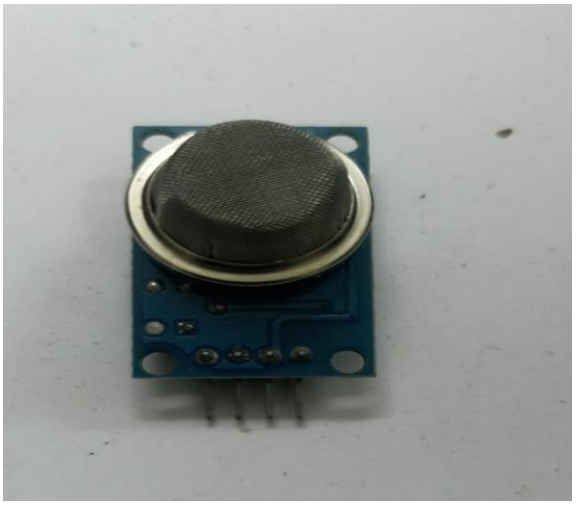

DHT11 Temperature and Humidity sensor

DHT11-it's a common temperature and humidity sensor. It's a low-cost sensor that provides pretty much accurate measurements.it has a range of 0°C to 50°C (32°F to 122°F) with an accuracy of ±2°C on temperature. For humidity it has the range of20% to 90% with an accuracy of ±5%.



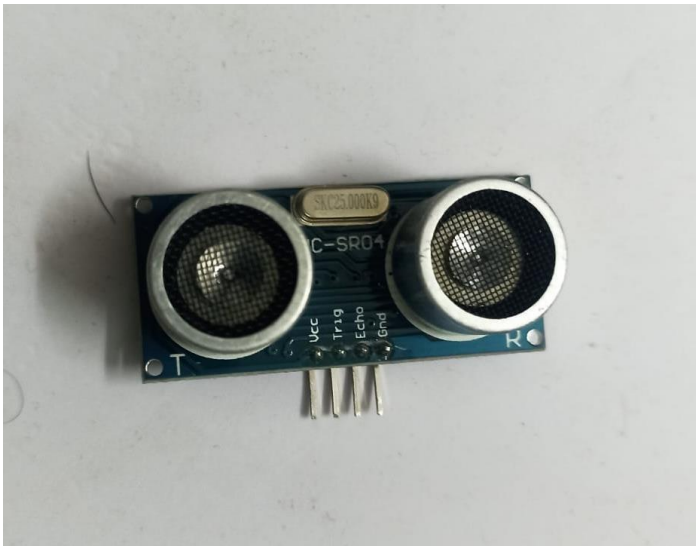
LCR Light Sensor

LDR -it's a light sensor. It changes its resistance based on the intensity of light falling on it.it is very sensitive to changes in light intensity. It can be used as  analog sensor than again we can use ADC to convert to digital. Its normal use is to automate lighting system.



MQ135 Gas sensor

MQ135 -its gas sensor that detect harmful gases in the air like ammonia, nitrogen oxides, benzene , smoke etc. It provides with analog output so we must use ADC to covert analog to digital. It consumes low power. It needs only 5 volt to operate and it’s a very affordable.

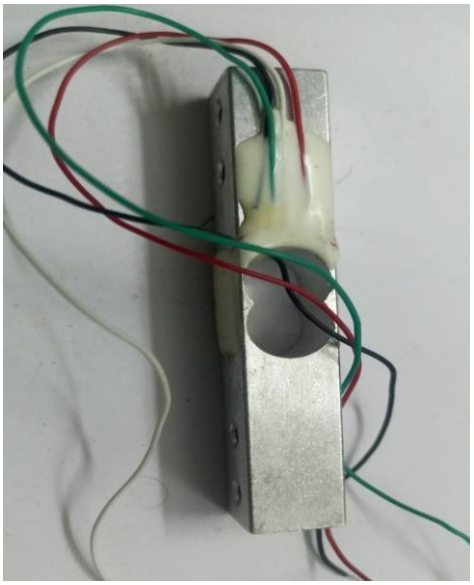



Ultrasonic Sensor

Ultrasonic sensors use sound waves at frequencies higher than the human audible range to detect the presence and distance of objects. The sensors emit ultrasonic waves typically in the range of 40 kHz to 200 kHz. It measure the time it takes for the sound waves to bounce back after hitting an object and calculate the time delay. This way sensor can calculate the distance or presence of a object.

Cooling fan

Cooling fan is using to regulate the temperature when it is become high. This module is used to decrease the temperature of the environment. It can operate in 5 to 9 volts. The maximum current it can tolerate is 1A the maximum rpm is 25000.



Module for weight sensor

Weight sensor modules come in various load capacities but the one we are using has only 1kg capacity. Weight sensor has one or more strain gauges. those are attached with a metallic structure. When there is a force or weight the strain gauges deform, and it causes a change in electrical resistance. This change in resistance is used to calculate the applied force or weight. Weight sensors can provide both digital and analog output. It depends on model or type. The requirement of power supply (volt) is also depending on the model. Usually, the range is 5-10 volt.

# Result

In the result, it is shown that, in cases of high or low temperature, the temperature sensor will monitor the change and give an indication. Then, in case any harmful gas, such as ammonia, is formed in the farm, it can be monitored with the help of the gas sensor and can be cleared by exhaust fan ventilation. The temperature, humidity, and air quality are uploaded to the cloud with the help of the wi-fi and displayed on the website. For the waste management system, there is an ultrasonic sensor and a weight sensor.

# Conclusion

The farm's environment is vitally important because it can affect the condition of the chickens. If the atmosphere is not well monitored, the problem can turn into a medical issue for the chickens. It is quite hard to monitor everything physically, especially at night times. Our model displays the utilization of IOT technology for layer chicken farming, keeping an eye on the farm and controlling the environment of the farm. The model sends data to the cloud and displays it on the website for easy monitoring and automating the hitter, cooler, ventilator fan, and west management system. So, the model is very effective in reducing manpower and saving time also. To manage the increasing demand for chicken and eggs in the world, farm owners need healthy chickens, and our model is highly effective for monitoring the good health of the chickens.

# References


[1] Archana M , Uma S , “Monitoring and controlling of poultry farm using IOT”, International Journal of Innovative Research in Computer and Communication Engineering, Vol. 6, Issue 4, April 2019.

[2] Zainal H. C. Sohl, Mohd H. Ismaill, “Development of automatic chicken feeder using Arduino Uno”, IEEE, Dec 2019.

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